SARDAR UMAIR NAWAZ KHAN

AI/ML & Backend Developer | Agentic AI Applications

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PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST-NUCES focused on AI-powered applications, machine learning, and backend engineering. Hands-on experience building LLM-powered automation workflows, REST APIs, and machine learning pipelines using Python, FastAPI, LangChain, n8n, and Scikit-learn. Experienced in integrating external APIs, relational databases, and Docker-based services.

TECHNICAL SKILLS

Programming Languages: Python, Modern C++ (C++17), TypeScript, JavaScript (ES6+), HTML5, CSS3, PHP, SQL, Bash

AI & Machine Learning: PyTorch, Scikit-learn, Machine Learning Workflows, Data Preprocessing, Model Fine-Tuning, Autonomous AI Agents, Agentic Workflows, LangChain, LangGraph, RAG Workflows, n8n Orchestration

Full-Stack & Backend: Next.js, React 18, FastAPI (Async Python), Node.js, Express.js, RESTful APIs, Asynchronous I/O (asyncpg), Vite

Databases & DevOps: PostgreSQL, MySQL, Relational Database Architecture, Docker, Linux, Git, GitHub, Postman

TECHNICAL PROJECTS

AI Appointment Assistant & Workflow Automation Engine [GitHub Code]

FastAPI, React 18, PostgreSQL, n8n, Google Gemini, Docker

- Built a multi-tenant appointment platform combining a React portal, asynchronous FastAPI APIs, PostgreSQL, and an n8n/Gemini booking workflow.
- Implemented availability validation, conflict-safe appointment booking, RFC-5545 calendar invitations, and a direct backend fallback path for booking continuity during AI or workflow downtime.

Autonomous AI Meeting Scheduling Agent [GitHub Code]

Python, n8n, Zoom API, Google Calendar API, Gmail API, Docker

- Built an n8n-based meeting orchestration workflow that processes webhook requests and creates Zoom video conferences.
- Integrated Zoom Server-to-Server OAuth, Google Calendar, and Gmail to synchronize events and send meeting confirmations.

Customer Segmentation via Hierarchical Clustering [GitHub Code]

Python, Scikit-learn, SciPy, Agglomerative Clustering, PCA, Pandas

- Aggregated 541,909 transaction line items into customer-level behavioral features across 4,372 accounts and applied two-component PCA, retaining 97.39% of variance.
- Applied Ward-linkage agglomerative clustering and dendrogram analysis to segment customers into three behavioral cohorts.

Financial Fraud Detection & Anomaly Classification [GitHub Code]

Python, Scikit-learn, Imbalanced-Learn, SMOTE, LinearSVC, Pandas

- Built a LinearSVC fraud-classification pipeline on 284,807 transactions and applied SMOTE only to the training split.
- Achieved 90.82% fraud recall (89/98 fraud cases) at 6.48% precision on the reported test set, highlighting the false-positive trade-off in an imbalanced dataset.

Full-Stack Hostel Management System [GitHub Code]

Node.js, Express.js, MySQL, Modular MVC, RESTful APIs

- Contributed to a team-built hostel management application using Node.js, Express, and MySQL, supporting room allocation, student records, staff administration, and fee tracking.
- Architected relational schema tables with foreign keys and implemented modular RESTful API endpoints for administrative workflows.

International Banking Management System [GitHub Code]

Modern C++17, OOP Architecture, File Stream Serialization, Data Persistence

- Developed a C++17 terminal-based banking management application using encapsulation, inheritance, and polymorphism.
- Implemented account authentication, defensive input validation, inter-account transfers, and CSV-based transaction persistence.

EDUCATION

FAST National University of Computer and Emerging Sciences (FAST-NUCES) Peshawar, Pakistan

Bachelor of Science in Computer Science (BSCS) **Aug 2024 – Jun 2028 (Expected)**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Operating Systems, Database Systems

Government Sadiq Abbas Postgraduate College Dera Nawab Sahib, Pakistan

HSSC (FSc Pre-Engineering) **2022 – 2024**

ADDITIONAL INFORMATION

Competitive Chess: Active rapid player (@umair-nawaz) with **4,500+ tactical puzzles solved** on Chess.com.